Faith Isdorah Onditi, Ph.D.

Curriculum Vitae

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Senior Malaria Research Scientist

Professional Summary

Global health scientist with over 10 years of experience in malaria research, vaccine development, and translational immunology. Expertise in non-human primate models, placental malaria, and preclinical vaccine and drug development. Proven ability to lead research projects from concept to publication, mentor trainees, and contribute to institutional research systems. Currently expanding into risk management and AI integration in health research.

Core Scientific and Research Interest

Placental malaria | Maternal–fetal immunology | Vaccine development | Translational infectious disease research | Non-human primate models | One Health | AI in health research systems

Professional Experience

Senior Research Scientist **Jan, 2026-Present**
Kenya Institute of Primate Research (KIPRE) — Department of Tropical & Infectious Diseases and One Health

- Lead infectious disease research within a One Health framework
- Chair, Risk Management Steering Committee (institutional policy development)
- Institutional AI Champion supporting early-stage AI integration in research systems
- Initiating stakeholder collaborations for AI capacity building

Research Fellow **2023–2025**
Laboratory of Malaria Immunology and Vaccinology, National Institute of Allergy and Infectious Diseases, National Institutes of Health (NIH)

- Optimized *Aotus nancymaae* chronic placental malaria model
- Conducted immunogenicity and functional antibody assays (ELISA, flow cytometry, binding inhibition)
- Contributed to PfSPZ and PVSPZ preclinical vaccine evaluation protocols
- Presented research at international conferences
- Reviewed scientific manuscripts and abstracts; judged NIH poster sessions
- Authored multiple conference abstracts

Senior Research Scientist **2016–2022**
Malaria Program, Department of Tropical and Infectious Diseases (TID), Kenya Institute of Primate Research (KIPRE), Nairobi Kenya

- Led malaria drug development and preclinical studies
- Secured research funding and built international collaborations
- Mentored students and supervised research staff
- Published peer-reviewed manuscripts
- Presented at international conferences (oral and poster)

Principal Investigator **2021**
Synthetic Biology (iGEM Program)

- Founded institutional synthetic biology program

- Led team to iGEM Bronze Medal
- Secured funding and developed environmental biosensor prototype
- Mentored young researchers in the field
- Organized and pioneered the first Synthetic Biology Workshop in Kenya

Research Scientist **2009–2016**

Malaria Program, Department of Tropical and Infectious Diseases (TID), Kenya Institute of Primate Research (KIPRE), Nairobi Kenya

- Led congenital malaria research
- Applied immunological and molecular techniques
- Presented findings at scientific conferences
- Mentored undergraduate and diploma students

Education and Credentials

- Ph.D. in Biochemistry, School of Medicine, University of Nairobi, Nairobi, Kenya: **2015**
- MSc. in Molecular Medicine, Institute of Tropical Medicine and Infectious Diseases (ITROMID), Jomo Kenyatta University of Agriculture and Technology (JKUAT), Nairobi, Kenya.: **2010**
- BSc. in Biochemistry and Molecular Biology, Jomo Kenyatta University of Agriculture and Technology (JKUAT), Nairobi, Kenya: **2005**

Awards

Best Student Oral Presentation Award at the AIBBC 2014 and 8th International Workshop on Approaches to Single-Cell Analysis - January 2014

Grants

- International Genetically Engineered Machine (iGEM) Backman Coulter Reagents Grant (Amount:
- $7,000)- September 2021
- iGEM Safety Grant - July 2021
- Elsevier Foundation Travel Grant for Early Career Women Scientists from Low or Low-Middle Income Countries to Present Research Finding at the 17th International Congress of Infectious Diseases (ICID) - December 2015
- Travel Award to attend the East African Regional Workshop on Immunology of Parasitic Diseases, in Morogoro, Tanzania - January 2010
- National Commission for Science, Technology, and Innovation (NACOSTI), Grant No. NCST/5/003/W69 (Amount $12,800) - May 2009,

Professional Development & Training

- **Good Clinical Practice Course (US FDA focus),** Collaborative Institutional Training Initiative, NIH- August 2025
- **Biomedical 101 Course,** Collaborative Institutional Training Initiative, NIH- August 2025
- **Translational Science Training Program (TSTP),** National Center for Advancing Translational Sciences (NCATS), NIH-May 2025
- **Responsible Conduct of Research,** Office of Intramural Training & Education, NIH – May 2025
- **Teaching Science Certificate Course,** NIH- April 2025
- **Management Course: Leadership and Management Training Program**, Office of Intramural Training & Education, NIH - March 2024
- **Statistical Analysis Research Data/Statistical Tutorial**, Center for Cancer Training, NIH - March 2024
- **Systematic Review and Meta-analysis Workshop**, Infectious Diseases Institute, Kampala, Uganda - September 2023
- **Virtual Workshop on Issues Emerging under the Convention on Biological Diversity (CBD)**, International Service for the Acquisition of Agribiotech Applications (ISAAA) - April 2022
- **Introduction to Linux, mapping & variant calling, R, Transcriptomic Data Analysis, and Visualization of Genomic Data**, African Bioinformatics Network (ABN) Summer school program - August 2021
- **Certificate Course in Research Administration**, Kenya Medical Research Institute (KEMRI), Nairobi, Kenya - June 2020

- **Certificate Course in Research Ethics for Institutional Ethics Review Committee**, Egerton University, Kenya - November 2019
- **Training on Foundation of Bioethics**, KEMRI Training Center, Nairobi, Kenya - September 2019
- **Introduction to Experimental Design and Analysis using GEN STAT Edition 15 and R**, KIPRE - October 2012
- **Working with Animals and Animal Vectors (Ticks) Under Containment Levels I, II and III**, International Livestock Research Institute (ILRI), Kenya - October 2012
- **Graduate, School of Leaders (SOL)**, Nairobi Lighthouse Church - December 2012
- **East African Regional Workshop on Immunology of Parasitic Diseases**, Morogoro, Tanzania - July 2010
- **UNICEF/UNDP/World Bank/WHO Special Program for Research and Training in Tropical Diseases (TDR) Workshop on Effective Project Planning in Biomedical Research**, Kenya Agricultural Research Institute – Trypanosomiasis Research Centre (KARI-TRC), Kenya - April 2008

Scientific & Professional Services

- **Chair, Risk Management Steering Committee-** 2026
- **Institutional AI Champion-** 2026
- **Invited member, International PfSPZ Consortium (iPfSPZ-c)-** 2025
- **Peer Review for Scientific Journals: PLOS One Journal, Biomedical and Pharmacology Journal,**
- **Judge**, NIH Postbac Poster Day - 2024
- **Reviewer for Crowd Pre-Print Peer Reviewer** - 2023
- **Judge**, NIH Hybrid Postbac Poster Day - 2023
- **Development of the National Museums of Kenya Intellectual Property Policy** - 2022
- **Development of COVID-19 Standard Operating Procedures (SOPs) for the Kenya Institute of Primate Research's COVID-19 Research and Testing Center** - 2020
- **Preclinical study proposal reviewer**, Kenya Institute of Primate Research, Institutional Scientific Ethics Review Committee (ISERC) - 2014-2022
- **Development of Intellectual Policy**, National Museums of Kenya, 2022
- **Development of COVID-19 Standard Operating Procedure,** KIPRE COVID-19 Research and Testing Center, 2020

Leadership Roles

- **Chairperson:** Risk Management Steering Committee, KIPRE, 2025
- **AI Champion:** KIPRE, 2026
- **Principal investigator,** ABSI team, iGEM - 2022
- **Team Leader,** Malaria Program, KIPRE - 2016-2022
- **Secretary, KIPRE,** Institutional Scientific Ethics Review Committee (ISERC) - 2016-2022
- **Cell Group Leader,** Nairobi Lighthouse Church, Kenya - 2012-2022
- **Vice Chairperson,** JKUAT- Alternative Degree Students Association (JKU_ADSA) - 2001-2022
- **School Prefect/Captain,** Aga Khan High School, Kenya - 2000-2019

Mentorship Roles

- Primary mentor to 1 PhD., 1MSc. and 7BSc. students
- Secondary mentor to 2PhD. and 2MSc. students

Affiliations

- **Member:** National Postdoc Association (NPA) - 2024
- **Participant,** Federal Malaria Research Network (FMRN) - 2024
- **Member,** The Society for Biomedical-biotechnology Research (SBBR) - Membership No. SBBR_078 - 2024
- **Member,** Communications and Outreach Committee, Genetics Society of American Society, Early Career Leadership Program - Membership No. 416818 - 2021 - Preset

Peer Reviewed Publications

1. Mogaka S., **Onditi F.**, Ogila K., Higashi C., Ozwara H. Antimalarial drug discovery and development: Assay system challenges and the translational relevance of Plasmodium knowlesi–baboon model.J Adv Biotechnol Exp Ther. 2026 Jan; 9(1): 52-66**.**
2. Mogaka S, Mulei I., Njoki P., Ogila K., Waihenya R., **Onditi F.**, Ozwara H. Antimalarial Efficacy and Safety of *Senna occidentalis* (L.) Link Root Extract in *Plasmodium berghei*-Infected BALB/c Mice. Biomedical Research International. Biomed Research International (2023), vol. 2023, article ID 8296195.
3. Mogaka S., Molu S., Kagasi E., Ogila K., Waihenya R., **Onditi F.**, Ozwara H. *Senna occidentalis* (L.) Link root extract inhibits *Plasmodium* growth *in vitro* and in mice. BMC Complementary Medicine and Therapies (2023), 71 (2023).
4. Nyirongo J., **Onditi F.**, Simbauni J., Kamau L., Mwangi V., Kagasi E., Ozwara H. Astemizole-Methylene Blue combination therapy reduces monotherapy adverse effects in Balb/C mice. Journal of Pharmaceutical Research International (2022) 34(53A):42-52
5. Nyirongo J., Kamau L., Simbauni J., Mwangi V., Kagasi E., **Onditi F.**, Ozwara H. Drug Recovery: Effect of Astemizole-Methylene Blue combination therapy against *Plasmodium* strains *in Vitro*, South Asian Journal of Parasitology (2020) 4(3):9-17
6. Taylor E., **Onditi F**., Maina N., Ozwara H. Immunization of mice with soluble lysate of interferon-gamma expressing *Plasmodium berghei* ANKA induces high IFN-γ production. Tropical Diseases, Travel Medicine and Vaccines (2017) 3:11
7. Mogaka S., Muriithi C., **Onditi, F**.I., Mumo R, Maina, N, Waihenya R., Ozwara, H.S. *Plasmodium berghei* is immunomodulated by transgenic mouse interferon-gamma leading to enhanced malaria protection in mice, Journal of Pharmacy and Biological Sciences,2015
8. **Onditi, F**.I., Nyamongo, W.O., Omwandho, C., Maina, N., Maloba, F., Farah, I.O., King, C., Moore, J., Ozwara, H.S. Parasite accumulation in placenta of non-immune baboons during *Plasmodium knowlesi* infection. Malaria Journal (2015) 14:118.

Published Conference Abstracts:

1. **Onditi F.,** Howard Jen., Nadakal S., Torano Holly., Doritchamou J., Spiegel ., Stevens R., Morrison R., Neal J., Raitin J., Bock K., Ouahes T., Orr-Gonzalez S., Alves D., Lambert L., Fried M., Duffy P: *Aotus nancymaae* as a Novel Non-human Primate Model of Chronic Placental Malaria. NIAID 18th Annual Fellows Workshop, 2024
2. **Onditi F.,** Renn., Lambert L., Ouahes T., Orr-Gonzalez S., Neal J., Alani N., Doritchamou J., Fried M., Duffy P: Impact of *in utero* malaria exposure on susceptibility of *Aotus nancymaae* offspring to *Plasmodium falciparum* parasitemia during early years of life**.** American Society of Tropical Medicine & Hygiene, 2024 Annual Meeting
3. **Onditi F.,** Lambert L., Ouahes T., OrrGonzalez S., Neal J., Chakravarty S., James E., Sim K., Stephen L. Hoffman S., Duffy P. First liver and blood stage infection of *Saimiri boliviensis* and *Saimiri sciureus* by inoculation of purified, cryopreserved Sanaria® *Plasmodium vivax* Sporozoite (PfSPZ), American Society of Tropical Medicine & Hygiene, 2023 Annual Meeting
4. **Onditi F.**, Howard J., Nadakal S., Torano H., Doritchamou J., Spiegel A., Morrison B., Neal J., Raitin J., Ouahes T, OrrGonzalez S., Lambert L., Fried M., Duffy P. *Aotus nancymaae* as a novel non-human primate model of chronic placental malaria. 9th Future of Malaria Research Symposium abstract book, 2023, pg. 10

5. Oduor J., Onkoba N., Maloba F, Arodi O., Nyachieo A., **Onditi F**. Safety and therapeutic efficacy of Staphylococcus aureus specific lytic phage against multidrug-resistant *S. aureus* (MDRSA) in BALB/c mice: A prospective study. International Journal of Infectious Diseases (2016) 45:108
6. **Onditi F.**, Omwandho C., Ozwara H. The baboon (*Papio anubis)-Plasmodium knowlesi* model of placental malaria. International Journal of Infectious Diseases (2016) 45:62
7. **Onditi F.**, Omwandho C., Maloba F., Christopher K., Moore M., Hastings Ozwara H, *In Utero* Sensitization of *Plasmodium knowlesi* in baboon infants., 2014, URI: http://repository.kyuc.ac.ke/123456789/166
8. **Onditi F.,** Onkoba N., Omwandho C., Ozwara H, Moore J., Baboon (*Papio anubis*) as a potential model for understanding immunological mechanism of placental malaria, 2012. URI: http://repository.kyuc.ac.ke/123456789/168

9. **Onditi F.**, Onkoba N., Omwandho C., Ozwara H., Moore J., Baboon *(Papio anubis*) as a model for developing future diagnostic tools for placental malaria in humans, 2012.URI: http://repository.kyuc.ac.ke/123456789/167
10. **Onditi** F., Onkoba N., Omwandho C., Maina N., Maloba F, Farah O., King C., Moore J., Ozwara H., Placental malaria pathology and fetal outcomes in baboon (*Papio anubis*) infected with *Plasmodium knowlesi* H strain, 2010. URI: http://hdl.handle.net/123456789/99.

Manuscripts in Preparation:

1. **Onditi F.**, Zaidi I., Lambert L., Ouahes T., Orr-Gonzalez S., Neal J., Morrison R., Chakravarty S, Natasha C, James E., Sim ., . Hoffman S., Duffy P.: First liver and blood stage infection of *Aotus griseimembra* with aseptic, purified, cryopreserved *Plasmodium falciparum* (NF54 strain) sporozoites.
2. **Onditi F.,** Howard J., Nadakal S., Torano H., Doritchamou J., Spiegel A ., Morrison R., Neal J., Raitin J., Bock K**.,** Ouahes T., Orr-Gonzalez S., Alves D., Lambert L., Thomas M., Fried M., Duffy P.: *Aotus nancymaae* as a Novel Non-human Primate Model of Chronic Placental Malaria.
3. **Onditi F.,** Lambert L., Ouahes T., OrrGonzalez S., Neal J., Chakravarty S., James E., Sim K., Stephen L. Hoffman S., Duffy P. First liver and blood stage infection of *Saimiri boliviensis* and *Saimiri sciureus* by inoculation of purified, cryopreserved Sanaria® *Plasmodium vivax* sporozoite (PfSPZ)

Conferences and Presentations/Talks

1. **Ethical Review of Animal Research and Its Role in Biomedical Research at KIPRE**, 2nd Joint annual Research Ethics Conference, Daystar University, Nairobi, Kenya, 2022
2. **Career talk for BSs. Medical Microbiology Final Year Students**, Organized by Jimo Kenyatta University Medical Microbiology Student Association (JKUMMA), 2022
3. **Virtual Workshop on Synthetic Biology as a Bioeconomy Prospect in Kenya (The How of Synthetic Biology)**, Institute of Primate Research, Nairobi, 2021.
4. **Importance of Animal Models in Understanding the Impact of Malaria in Pregnancy in the Era of COVID-19**, the 5th Africa International Biotechnology and Biomedical Conference (AIBBC), Kisumu, Kenya, 2021
5. **Role of Non-Human Primates in Understanding the Dynamics of Malaria in Pregnancy: The Baboon Model of Placental Malaria**, the International Conference on Health (ICH), Kenyatta National Hospital, Nairobi, Kenya. 2018
6. **The baboon Model of Placental Malaria is Ideal for Experimental Therapeutic Management of Malaria in Pregnancy**, the 3rd AIBBC 2017, Nairobi, Kenya, 2017
7. **The baboon (*Papio anubis*)-*Plasmodium knowlesi* Model of Placental Malaria**, the 17th International Congress of Infectious Diseases (ICID), Hyderabad, India, 2016
8. **Baboon infants are protected *in utero* against *P. knowlesi* infection, Joint Meeting of the 1st Africa International Biotechnology & Biomedical Conference and the 8th International Workshop on Approaches to Single-Cell Analysis**, Safari Park Hotel, Nairobi, Kenya, September 2014.

References: Available upon request